

## Multiple Choice (10 marks)

1. You've made a scientific theory that there is an attractive force between all objects. When will you know the theory is correct?
  - (a) When the theory can predict future observations.
  - (b) As soon as you propose the theory.
  - (c) When your theory becomes a law.
  - (d) You can never prove your theory to be correct only "yet to be proven wrong".
  - (e) When the scientific community accepts it as a fact.
2. According to Coulomb, a pair of charged particles placed twice as close to each other experience a force
  - (a) four times as strong
  - (b) half as strong
  - (c) one-fourth as strong
  - (d) inversely proportional to the distance
  - (e) twice as strong
3. A particle of charge  $1.8 \times 10^{-5} \text{ C}$  is placed in an electric field of  $2.0 \times 10^5 \text{ N/C}$ . The force on the particle is
  4.  $2.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
  5.  $2.8 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
  6.  $3.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
  7.  $1.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
  8.  $1.5 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$

Which statement about an atom is not true:

1. A neutral atom always has equal numbers of electrons and protons.
2. The electrons can only orbit at particular energy levels.
3. Electrons and protons in a neutral atom are always equal in number.
4. Electrons can orbit at any energy level.
5. A neutral atom always has equal numbers of neutrons and protons.

The plates of a capacitor are charged to a potential difference of 5 V. If the capacitance is 2 mF, what is the charge on the plates?

1. 2 C
2. 0.02 C

3. 0.02 mC
4. 0.005 C
5. 0.1 C

## True / False (10 marks)

*Answer True or False*

6. True or False: The cost to run a motor that draws 5 amperes from a 120-volt line for 2 hours is 15
7. True or False: For a particle moving with constant velocity along a cardioid, the radial component
8. True or False: The Mars Exploration Rover Spirit is tilted towards the north because it is in the so
9. True or False: The electrostatic force between two positive charges of 100C and 200C separated
10. True or False: The electric field produced by a finite-length charged rod decreases as the inverse

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Given point charges of +6.0 C at  $x=8.0$  m and -4.0 C at  $x=16.0$  m, determine the required charge at  $x=24.0$  m for a charge at the origin to experience no net electrostatic force. Explain your reasoning and calculations.
12. Given a 1.4447-gm sample of coal burned in an oxygen-bomb calorimeter, analyze the temperature rise and the system's water equivalent to calculate the heating value of the coal. Discuss the implications of your findings.

*End of Paper*